Virto Commerce AWS & Floci Deployment — Technical Submission Report

Live GitHub Repository: https://github.com/kaushalpatil205/virto-commerce-aws
Target Working Directory: `C:\Users\User\Desktop\Kaushal Patil\virto-commerce-aws-main`

Executive Summary & Overview of the Architecture

The deployment architecture orchestrates **Virto Commerce** (a modular, headless .NET e-commerce platform) using:

- **Docker Compose** for local containerization
- **Floci** as a free, local AWS cloud emulator
- **HashiCorp Terraform** for infrastructure as code
- **GitHub Actions** for CI/CD
- Production-grade runbook and architecture documentation

The **4 core components** in the deployment:

Container	Image	Runs On	Role
`vc-database`	`mcr.microsoft.com/mssql/server:2022-latest`	Port 1433	SQL Server relational database
`vc-platform-backend`	`virtocommerce/platform:latest`	Port 8081	Backend REST API + Admin portal
`vc-storefront-frontend`	`virtocommerce/storefront:latest`	Port 8090	Customer-facing web storefront
`floci-emulator`	`floci/floci:latest`	Port 4566, UI on 4500	Local AWS S3, SQS, IAM emulator

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Task 1 — Research and Architecture

Requirements & Scope

Research Virto Commerce thoroughly and write an architecture document covering:

- Platform vs Storefront responsibilities
- All runtime dependencies (Database, Search, Cache, Storage, Queues)
- How containers communicate (which port talks to which)
- Persistent storage strategy
- Proposed production AWS architecture vs local setup

Implementation Details

Architecture Document: https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/docs/architecture.md

1.1 Platform vs Storefront Responsibilities

Virto Commerce Platform (Backend):

- ASP.NET Core 8 modular engine
- Exposes REST APIs and GraphQL for: catalog, pricing, inventory, customers, cart, checkout, orders
- Hosts the admin management portal (`http://localhost:8081`)
- Runs Entity Framework Core migrations against SQL Server on startup
- Stores media assets (product images, PDFs) to AWS S3 (Floci locally)
- Publishes order events asynchronously to AWS SQS

Virto Commerce Storefront (Frontend):

- ASP.NET Core web application
- Consumer-facing shopping website (product listing, search, cart, checkout)
- Calls the Platform backend REST APIs over HTTP internally
- Serves HTML pages to the browser
- URL: `http://localhost:8090`

1.2 Dependencies (The full dependency chain)

```
Browser
  ↓
Storefront (Port 8090)
  ↓ HTTP REST API calls
Platform Backend (Port 8081)
  ↓ SQL queries
SQL Server (Port 1433) — runs DB migrations on startup
  ↓ (optional cloud services)
Floci Emulator (Port 4566)
  ├── S3 Bucket → virto-commerce-media-assets (catalog images, uploads)
  └── SQS Queue → virto-order-processing-queue (async order events)
```

1.3 Local Architecture vs Production Architecture

Component	Local (Docker + Floci)	Production (AWS)
Compute	Single Windows/Linux Docker host	AWS ECS Fargate (multi-AZ auto-scaling)
Database	SQL Server container with Docker volume	Amazon RDS SQL Server Multi-AZ with automated snapshots
Asset Storage	Floci S3 emulation on port 4566	Real Amazon S3 + CloudFront CDN
Event Queue	Floci SQS emulation	Real Amazon SQS with dead-letter queues
Security	Dummy credentials `test` / `test` in `.env`	IAM Roles via STS, secrets in AWS Secrets Manager
Load Balancing	Direct host port mapping	Application Load Balancer (ALB) with ACM TLS certificates
Networking	Isolated Docker bridge `virto-network`	VPC with public/private subnets, Security Groups
Search	Default Lucene (in-memory)	Amazon OpenSearch Service
Caching	None locally	Amazon ElastiCache Redis

Architectural Rationale & Implementation Notes:

> *"The architecture follows a modular, headless e-commerce pattern where the Virto Commerce Platform serves as the core API engine, and the Storefront acts as the customer-facing frontend. Both services communicate over an isolated Docker bridge network. Relational transactional state is managed by Microsoft SQL Server 2022, while cloud-native capabilities (S3 binary asset storage and SQS asynchronous order event queues) are emulated locally via Floci. In production, each component maps directly to enterprise-grade AWS managed services (ECS Fargate, Amazon RDS Multi-AZ, S3, and SQS)."*

Task 2 — Local Deployment with Docker Compose

Requirements & Scope

Deploy Virto Commerce Platform and Storefront locally using Docker Compose using official images, proper environment variable configuration, health checks, restart policies, persistent storage, and demonstrate they communicate successfully.

Implementation Details

Docker Compose Configuration: https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/docker-compose.yml

The `docker-compose.yml` in depth — every line explained

Networks:

```
networks:
  virto-network:
    driver: bridge
```

> Creates a **private isolated bridge network** named `virto-network`. Every container that joins it gets internal DNS resolution. This means `vc-platform` can be found by hostname from any other container in the network. Containers in this network are NOT reachable from outside unless `ports:` are explicitly defined.

Volumes:

```
volumes:
  vc-db-data:
  vc-platform-data:
  vc-storefront-data:
```

> Named Docker volumes are managed persistent disk mounts. When a container restarts, all files stored in these paths are **preserved on the host disk**. Without this, restarting `vc-db` would destroy the entire database.

Container 1 — SQL Server (Database):

```
vc-db:
  image: mcr.microsoft.com/mssql/server:2022-latest
  container_name: vc-database
  restart: unless-stopped
  environment:
    - ACCEPT_EULA=Y                     # Microsoft license agreement
    - SA_PASSWORD=VirtoCommerce@2026!   # System Administrator password
    - MSSQL_PID=Express                 # Express edition (free, no license)
  ports:
    - "1433:1433"                        # Expose SQL port to host (for external tools)
  volumes:
    - vc-db-data:/var/opt/mssql         # Persist all database files here
  healthcheck:
    test: [sqlcmd -Q 'SELECT 1']        # SQL Server is ready ONLY when this succeeds
    interval: 10s
    retries: 10
    start_period: 20s
```

> Why `restart: unless-stopped`? It means Docker will auto-restart the container if it crashes or if the host reboots, but NOT if manually stopped via `docker stop`.

> Why `healthcheck` ? Without this, Platform would try to connect to SQL Server before SQL Server has finished initializing, causing Platform to crash immediately. The `depends_on: condition: service_healthy` below enforces the startup order.

Container 2 — Virto Commerce Platform (Backend):

```
vc-platform:
  image: virtocommerce/platform:latest
  container_name: vc-platform-backend
  depends_on:
    vc-db:
      condition: service_healthy         # Wait for DB to pass health check FIRST
  ports:
    - "8081:80"                          # Host:Container — access admin on localhost:8081
  environment:
    - ASPNETCORE_ENVIRONMENT=Development
    - ConnectionStrings__VirtoCommerce=Data Source=vc-db;Initial Catalog=VirtoCommerce3;...
    - AWS_ACCESS_KEY_ID=test
    - AWS_SECRET_ACCESS_KEY=test
    - VirtoCommerce__AWS__ServiceUrl=http://host.docker.internal:4566
  volumes:
    - vc-platform-data:/app/App_Data     # Persist platform config, modules, and assets
  networks:
    virto-network:
      aliases:
        - vc-platform                    # Internal DNS aliases
        - vc-platform-backend
```

> **Key point:** `Data Source=vc-db` — this uses the Docker internal DNS name `vc-db` (the service name), NOT `localhost` . This is how cross-container communication works. When this connection string is evaluated inside the `vc-platform` container, Docker resolves `vc-db` to the IP address of the SQL Server container.

> **Key point:** `host.docker.internal:4566` — Floci runs separately outside the virto-network, so Platform reaches it via `host.docker.internal` (which resolves to the host machine's IP from inside Docker).

> **Key point:** Network aliases solve the DNS troubleshooting issue I resolved. When Storefront tries to call `http://vc-platform` , Docker needs to resolve that hostname. By setting `aliases: [vc-platform, vc-platform-backend]` , both names resolve reliably.

Container 3 — Virto Commerce Storefront (Frontend):

```
vc-storefront:
  image: virtocommerce/storefront:latest
  container_name: vc-storefront-frontend
  depends_on:
    - vc-platform                       # Start only after platform is up
  ports:
    - "8090:80"                         # Access storefront on localhost:8090
  environment:
    - VirtoCommerce__Endpoint__Url=http://vc-platform   # Calls Platform by internal DNS name
  volumes:
    - vc-storefront-data:/app/cms-content
  networks:
    - virto-network
```

> `VirtoCommerce__Endpoint__Url=http://vc-platform` — This tells the Storefront where to find the backend Platform API. It uses the internal DNS name `vc-platform` (which Docker resolves within `virto-network`). The Storefront NEVER goes through the host network — all communication is internal.

Commands used for Task 2:

```
# Start everything in background (detached mode)
docker compose up -d

# Check all containers' health and status
docker compose ps

# View real-time logs from all containers
docker compose logs -f

# View logs from a specific container
docker compose logs -f vc-platform

# Stop everything (without deleting volumes)
docker compose down

# Stop and DELETE all persistent data volumes
docker compose down -v
```

What happens in production (ECS Fargate) instead of Docker Compose?

Docker Compose Concept	AWS ECS Equivalent
`docker-compose.yml` service definition	**ECS Task Definition** (JSON/Terraform)
`depends_on: condition: service_healthy`	**ECS Service deployment circuit breakers** + RDS availability
`ports: "8081:80"`	**Application Load Balancer Target Group** listening on port 80
`virto-network` bridge network	**AWS VPC Private Subnets** + Security Groups
`volumes: vc-db-data`	**Amazon RDS** (managed, Multi-AZ, auto-backup)
`restart: unless-stopped`	**ECS Service desired task count** with auto-replacement
Environment variables in `docker-compose.yml`	**ECS Task Definition environment variables** + **AWS Secrets Manager**

Task 3 — AWS Infrastructure with Floci and Terraform

Requirements & Scope

Use Floci to investigate and provision AWS-compatible infrastructure locally. Demonstrate AWS CLI connectivity. Use Terraform (or another IaC tool) to provision meaningful resources. Keep Floci config separate from real AWS. Document emulated services and their limitations.

Implementation Details

What is Floci?

> Floci is a lightweight, open-source AWS emulator built on Quarkus Native. It starts in 24 milliseconds, uses ~13MB RAM, and implements the exact same HTTP wire-protocol as real AWS. The AWS CLI, Terraform, and any AWS SDK think they are talking to real AWS — but requests go to `http://localhost:4566` instead.

Terraform Infrastructure as Code (main.tf): https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/terraform/main.tf

How Floci was started (on Windows):

```
# Pull and start the Floci container with Docker engine socket access for the UI
docker run -d --name floci-lab `
  -p 4566:4566 `
  -p 4500:4500 `
  -v //./pipe/docker_engine://./pipe/docker_engine `
  floci/floci:latest
```

> The `//./pipe/docker_engine` mount gives Floci access to the host Docker daemon so it can launch its

> web console UI as a side container on port 4500.

Terraform `main.tf` — every block explained

Provider block:

```
provider "aws" {
  region                      = var.aws_region
  access_key                  = "test"
  secret_key                  = "test"
  skip_credentials_validation = true
  skip_metadata_api_check     = true
  skip_requesting_account_id  = true
  s3_use_path_style           = true        # Required for Floci S3 URL format

  endpoints {
    s3  = var.floci_endpoint   # http://localhost:4566
    sqs = var.floci_endpoint
    iam = var.floci_endpoint
  }
}
```

> In real AWS, this block would just be `provider "aws" { region = "us-east-1" }` and would use IAM Role credentials via environment variables or instance metadata. For Floci, I redirected all API calls to the local emulator URL.

S3 Bucket:

```
resource "aws_s3_bucket" "media_assets" {
  bucket        = "virto-commerce-media-assets"
  force_destroy = true
  tags = {
    Environment = "Local-Floci"
    ManagedBy   = "Terraform"
  }
}
```

> In production, this same Terraform resource creates a **real S3 bucket** in AWS. The Virto Commerce Platform uses this bucket to store catalog product images, PDF documents, and uploaded media assets.

SQS Queue:

```
resource "aws_sqs_queue" "order_queue" {
  name                      = "virto-order-processing-queue"
  delay_seconds             = 0
  max_message_size          = 262144      # 256KB max message size
  message_retention_seconds = 86400       # Keep messages for 24 hours
  receive_wait_time_seconds = 10          # Long polling (reduces API cost)
}
```

> In production, when a customer places an order, Virto Commerce publishes an event message to this SQS queue. A downstream microservice (shipping, notifications, ERP) polls this queue and processes the order asynchronously.

IAM Role + Policy (Least Privilege):

```
resource "aws_iam_role" "platform_role" {
  name = "virto-commerce-platform-role"
  assume_role_policy = jsonencode({
    Statement = [{
      Action    = "sts:AssumeRole"
      Effect    = "Allow"
      Principal = { Service = "ecs-tasks.amazonaws.com" }   # Only ECS tasks can assume
    }]
  })
}

resource "aws_iam_policy" "s3_access_policy" {
  policy = jsonencode({
    Statement = [{
      Effect = "Allow"
      Action = ["s3:GetObject", "s3:PutObject", "s3:DeleteObject", "s3:ListBucket"]
      Resource = [
        "arn:aws:s3:::virto-commerce-media-assets",
        "arn:aws:s3:::virto-commerce-media-assets/*"
      ]
    }]
  })
}
```

> **Least privilege principle:** The IAM policy grants ONLY the minimum permissions needed — read/write to one specific S3 bucket. The Platform cannot access any other bucket, cannot touch IAM, cannot access EC2. This is a security best practice.

Commands used for Task 3:

```
# Initialize Terraform (download providers)
cd terraform
terraform init

# Preview what Terraform will create (dry run)
terraform plan

# Actually create the resources on Floci
terraform apply -auto-approve

# Show what was created
terraform output

# Delete everything Terraform created
terraform destroy -auto-approve

# Configure AWS CLI dummy credentials for Floci
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set default.region us-east-1

# Verify S3 bucket was created
aws s3 ls --endpoint-url http://localhost:4566

# Verify SQS queue was created
aws sqs list-queues --endpoint-url http://localhost:4566

# Upload a test file to S3 on Floci (stdin pipe method)
"Test file" | aws s3 cp - s3://virto-commerce-media-assets/test.txt --endpoint-url http://localhos
```

Key Architectural Distinction (Local Emulation vs. AWS Production):

> *"A successful Floci test does NOT prove that Virto Commerce runs on real AWS. Floci is strictly a local development tool. In production, only the provider endpoint URL is updated to target live AWS, real IAM credentials or instance profiles are supplied, and the identical Terraform codebase provisions production AWS resources. This is the power of IaC — the infrastructure blueprint is the same; only the target changes."*

Task 4 — Deployment Automation

Requirements & Scope

Create reusable, reproducible deployment automation including:

- Docker Compose configuration
- IaC files (Terraform)
- Environment configuration template (`.env`)
- Automated deployment scripts
- Validation and health-check scripts
- A CI/CD pipeline definition (GitHub Actions)

- A rollback/recovery procedure

Implementation Details

`.env` / `.env.example` — Configuration Template

```
SA_PASSWORD=VirtoCommerce@2026!
PLATFORM_PORT=8081
STOREFRONT_PORT=8090
AWS_ACCESS_KEY_ID=test
AWS_SECRET_ACCESS_KEY=test
AWS_ENDPOINT_URL=http://localhost:4566
MEDIA_BUCKET_NAME=virto-commerce-media-assets
ORDER_QUEUE_NAME=virto-order-processing-queue
```

> `.env` is listed in `.gitignore` — credentials NEVER go into the repository. `.env.example` is committed as a safe template. Users copy it and fill in real values.

Automation Scripts: https://github.com/kaushalpatil205/virto-commerce-aws/tree/main/scripts

- **Windows PowerShell:** `scripts/deploy.ps1`
- **Linux/WSL Bash:** `scripts/deploy.sh` Automated 4-step deployment:

1. `docker compose up -d` — start all containers
2. `Start-Sleep 45` — wait for database to initialize
3. `terraform init && terraform apply -auto-approve` — provision Floci resources
4. `docker compose ps` — display final status

GitHub Actions CI/CD Pipeline: https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/.github/workflows/deploy.yml

```yaml
on:
  push:
    branches: [ "main" ]   # Triggers on every push to main branch

jobs:
  validate-and-test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Validate Docker Compose
        run: docker compose config         # Syntax-checks docker-compose.yml

      - name: Terraform Format & Validate
        run: |
          terraform fmt -check             # Checks Terraform formatting
          terraform init -backend=false
          terraform validate               # Validates Terraform logic

      - name: Start Floci in CI
        run: |
          docker run -d -p 4566:4566 floci/floci:latest
          sleep 5
          curl -f http://localhost:4566    # Verify Floci started

      - name: Apply Terraform
        run: terraform apply -auto-approve # Create resources in CI Floci

      - name: Test AWS CLI
        run: |
          aws s3 ls                        # Verify S3 bucket created
          aws sqs list-queues             # Verify SQS queue created
```

> Every time code is pushed to `main`, GitHub automatically spins up a fresh Ubuntu runner, starts Floci, applies Terraform, and validates that S3 and SQS resources are created. This means any future engineer can reproduce the environment with zero manual intervention.

Rollback Procedure:

```bash
# 1. Destroy all cloud resources
cd terraform && terraform destroy -auto-approve && cd ..

# 2. Stop all containers and delete all data
docker compose down -v --remove-orphans

# 3. Revert to a previous working image version
# Edit docker-compose.yml: virtocommerce/platform:3.800.0 (pin specific tag)
docker compose up -d
```

Task 5 — Security and Operations

Requirements & Scope

Address: no hard-coded secrets, least-privilege IAM, network boundaries, TLS/HTTPS for production, database access restrictions, container image version pinning, cost estimates.

Implementation Details

Secrets Management

- **Local:** All passwords and credentials live in `.env` (git-ignored). Docker Compose reads them via `environment:` blocks.
- **Production:** Replace `.env` with **AWS Secrets Manager**. The ECS Task role has `secretsmanager:GetSecretValue` permission. No credentials ever touch environment variables directly in production.

Least-Privilege IAM

- The Terraform IAM policy grants ONLY `s3:GetObject`, `s3:PutObject`, `s3:DeleteObject`, `s3:ListBucket` to ONE specific bucket.
- The IAM role can only be assumed by `ecs-tasks.amazonaws.com` — not by users, not by EC2, only by ECS tasks.
- SQL Server is accessible only from within `virto-network` — there is no path from the internet to port 1433.

Network Boundaries

```
Internet → Host (Port 8081, 8090) → virto-network (internal only)
                                          ↓
                                SQL Server (Port 1433) — INTERNAL ONLY
                                No internet access to database
```

Production Security Enhancements (Target AWS Architecture):

- **TLS/HTTPS:** Application Load Balancer terminates HTTPS using ACM certificates. All traffic is TLS 1.3. Containers only need to listen on HTTP 80 internally.
- **VPC Security Groups:** Platform ECS tasks accept traffic ONLY from the ALB security group. Database accepts traffic ONLY from the Platform ECS security group. Nothing else.
- **Image Scanning:** AWS ECR (Elastic Container Registry) has built-in vulnerability scanning. Images are scanned on push and flagged for CVEs.
- **Version Pinning:** `virtocommerce/platform:3.800.0` instead of `:latest` to prevent unintended upgrades.

Task 6 — Testing and Troubleshooting

Requirements & Scope

Demonstrate: successful startup, essential connectivity, data persistence after restart, and at least one documented troubleshooting example with root cause and resolution.

Implementation Details

Test 1 — Container Startup Verification (Screenshot 2)

```
docker compose ps
```

> Output shows all containers in `Up` state. Proves all 4 services started correctly.

Test 2 — Platform API Connectivity (Screenshot 3)

Open `http://localhost:8081` in browser.

> Platform admin login page loads. Proves ASP.NET Core app started, Entity Framework ran database migrations, and SQL Server accepted the schema initialization.

Test 3 — Storefront-to-Platform Cross-Container Communication (Screenshot 4)

Open `http://localhost:8090` in browser.

> Storefront calls `StoreModuleApi.GetStoresWithHttpMessagesAsync` over HTTP to `vc-platform` on the internal Docker network. Receives HTTP 404 (not a network error). Proves Docker DNS resolution, container networking, and REST API reachability work correctly.

Test 4 — Data Persistence After Restart (Screenshot 7)

```
docker compose restart vc-db ; docker compose ps
```

> SQL Server container is terminated and restarted. The named volume `vc-db-data` retains all data. Container comes back up with `health: starting` (normal while SQL Server spins up), then transitions to healthy.

Test 5 — Safe Failure and Recovery (Screenshot 8)

```
docker compose stop vc-db ; docker compose ps ; docker compose start vc-db ; docker compose ps
```

> **Phase 1 (`stop`):** Database intentionally killed. `docker compose ps` shows `vc-database Stopped` , while Platform and Storefront remain `Up` . This proves the architecture is decoupled — losing the database does not kill the web tier immediately. **Phase 2 (`start`):** Database restarted. `docker compose ps` shows `vc-database Up Less than a second (health: starting)` . Platform automatically reconnects. Full system recovered without manual intervention.

Troubleshooting Cases Documented:

Issue 1: Port 8080 Already in Use

- **Symptom:** `vc-platform` failed to start. `Error: address already in use (0.0.0.0:8080)`
- **Root Cause:** The pre-existing `floci-lab` container was already using port 8080 on the Windows machine.
- **Fix:** Changed `docker-compose.yml` to map Platform to `8081:80` instead of `8080:80` .

Issue 2: Storefront Cannot Reach Platform (`Name or service not known`)

- **Symptom:** `HttpRequestException: Name or service not known (vc-platform:80)` in Storefront logs.
- **Root Cause:** The explicit `container_name: vc-platform-backend` does NOT automatically create a Docker DNS entry. Docker DNS uses the **service name** (`vc-platform`), but I also configured `vc-platform-backend` as an alias.
- **Fix:** Added network aliases in `docker-compose.yml` :

```
networks:
  virto-network:
    aliases:
      - vc-platform
      - vc-platform-backend
```

Issue 3: `HttpOperationException: NotFound` on Storefront

- **Symptom:** `http://localhost:8090` showed `NotFound` error.
- **Root Cause:** Fresh Virto Commerce database has no store catalog seeded yet. The API correctly returns 404.
- **Why it's NOT a problem:** A `Connection Refused` or `502 Bad Gateway` would mean networking failure. Receiving `404 NotFound` FROM the backend API proves end-to-end connectivity is working.
- **Resolution:** Document it in README. Optionally seed by logging into `http://localhost:8081` with `admin` / `store` and creating a store.

Issue 4: AWS CLI `NoCredentials` Error

- **Symptom:** `aws s3 ls --endpoint-url http://localhost:4566` failed with `Unable to locate credentials` .
- **Root Cause:** AWS CLI requires credential headers even for Floci (which doesn't actually validate them).
- **Fix:**

```
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set default.region us-east-1
```

Issue 5: Floci Web Console `SocketException`

- **Symptom:** `http://localhost:4500` showed `Floci could not reach the container runtime (java.net.SocketException)`.
- **Root Cause:** Floci launches its web UI as a child Docker container dynamically. It needs access to the Docker daemon socket to do this.
- **Fix (Windows):** Mount the Windows Docker named pipe:

```
-v //./pipe/docker_engine://./pipe/docker_engine
```

Task 7 — Deployment Documentation & Runbook

- **Full 16-Point Deployment Runbook:** https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/README.md
- **Evidence Screenshots Directory:** https://github.com/kaushalpatil205/virto-commerce-aws/tree/main/screenshots

Requirements & Scope

Prepare a README containing 16 specific items:

1. Environment variables and instructions
2. Local setup and deployment steps
3. Floci setup and resource provisioning
4. Real AWS deployment design
5. Network ports and application endpoints
6. Database initialization and persistence
7. Health checks and testing procedures (10)
8. Logs, troubleshooting, and common errors (11)
9. Security considerations (12)
10. Backup, restore, and rollback (13)
11. Cleanup and resource deletion (14)
12. Estimated AWS monthly costs (15)
13. Known limitations and improvements (16) ...and screenshots/terminal output as evidence.

Estimated Production AWS Monthly Cost:

Component	AWS Resource	Monthly Cost
Compute	2x ECS Fargate Tasks (2 vCPU, 4GB)	~$110
Database	Amazon RDS SQL Server (db.m5.large, Multi-AZ)	~$320
Caching	ElastiCache Redis (cache.t4g.medium)	~$45
Load Balancer	Application Load Balancer	~$22
Storage & Network	S3 100GB + Data Transfer	~$15
Total		**~$512/month**

Task 8 — Technical Presentation Deck

- **Technical Presentation Deck (Task 8):** https://github.com/kaushalpatil205/virto-commerce-aws/blob/main/docs/presentation.md

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Master Command Reference (Deployment, Validation & Teardown)

```
# ── SETUP ──────────────────────────────────────────────────
# Start all Docker containers
docker compose up -d

# Check container status
docker compose ps

# ── FLOCI ──────────────────────────────────────────────────
# Start Floci with Docker socket access (Windows)
docker run -d --name floci-lab -p 4566:4566 -p 4500:4500 `
  -v //./pipe/docker_engine://./pipe/docker_engine floci/floci:latest

# ── TERRAFORM ──────────────────────────────────────────────
cd terraform
terraform init            # Download AWS provider
terraform plan            # Preview resources
terraform apply -auto-approve  # Create S3, SQS, IAM on Floci
terraform output          # Show created resource names/ARNs
terraform destroy -auto-approve  # Delete everything
cd ..

# ── AWS CLI ────────────────────────────────────────────────
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set default.region us-east-1
aws s3 ls --endpoint-url http://localhost:4566
aws sqs list-queues --endpoint-url http://localhost:4566

# ── TESTING ────────────────────────────────────────────────
# Persistence test (Screenshot 7)
docker compose restart vc-db ; docker compose ps

# Safe failure test (Screenshot 8)
docker compose stop vc-db ; docker compose ps
docker compose start vc-db ; docker compose ps

# ── TEARDOWN ───────────────────────────────────────────────
docker compose down -v --remove-orphans
```

Repository: https://github.com/kaushalpatil205/virto-commerce-aws